

Pranav Waghanna

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Education

University of Rochester, Master of Science in Computer Science	Aug 2024 – Dec 2025
Pune Institute of Computer Technology, Bachelor in Information Technology	Aug 2020 – Jun 2024

Experience

Fullstack Developer - Freelance Contract, Radha-Cottage	March 2026 – Present
• Architected a high-performance, responsive frontend for a heritage hospitality property using Typescript, HTML, CSS, Next.js and React, utilizing GSAP and Lenis for advanced scroll-triggered animations that deliver a narrative-driven UX while ensuring intuitive accessibility for a diverse demographic. • Built a secure, custom Content Management System (CMS) to dynamically manage events, booking packages, blogs, and media galleries, designing efficient MongoDB data models and integrating Cloudflare R2 for high-speed, global media delivery. • Engineered an interactive layered web loader that responds to precise navigation triggers, simultaneously optimizing core web vitals and implementing comprehensive SEO strategies across all routing structures to maximize organic discoverability.	
Backend Developer - Freelance Contract, Brydge	Feb 2026 – Present
• Developed responsive, high-performance frontend user interfaces for a social platform using ReactJS, HTML and modern CSS, ensuring cross-browser compatibility and sub-2 second mobile load times. • Designed scalable RESTful APIs and backend services using Node.js, Express, and PostgreSQL, while implementing an automated CI/CD pipeline via GitHub for rapid, reliable deployments. • Integrated Github Copilot and Claude into the development workflow, utilizing prompt engineering to optimize platform discoverability and system generation, directly contributing to Rs. 2L+ initial revenue. • Engineered a secure booking system with validation and QR-based UPI payments, along with automated email notifications, improving booking completion rates by 20%.	

Projects

LogStream Engine: High Performance Data Ingestion System (View Project)	March 2026 – May 2026
• Architected a high-performance, append-only data ingestion system from scratch in Rust using Tokio async I/O, achieving a throughput of over 60,000 messages/sec for continuous telemetry streams. • Designed a custom length-prefixed binary TCP protocol and implemented bounded concurrency with semaphore backpressure, eliminating HTTP framing overhead and ensuring zero message loss under burst loads. • Built a log-structured storage engine featuring an in-memory index for O(1) reads and 256MB segment rotation, deploying the containerized system via Docker with real-time Prometheus monitoring.	
eBPF System and File Access Monitor (View Project)	Nov 2025 – Jan 2026
• Developed a security tool on Linux that watches how programs access files and the network, using C, extended Berkeley Packet Filter(eBPF), Python and TCP/IP stack concepts to detect unusual activity in real time. • Implemented Linux internals with eBPF (BCC) in a Bodhi Linux virtual machine to trace syscalls, capturing 14K+ events in 32 seconds (447 events/sec) with zero loss for high-performance monitoring.	
Houdini: VFS-layer Rootkit for FreeBSD (View Project)	Sept 2025 – Dec 2025
• Built a stealth system program using C/C++ on a Unix/Linux-like system (FreeBSD) that hides files and spoofs system data, leveraging kernel-level development concepts to evade standard system visibility. • Used Unix internals and kernel-level development to hook VFS-layer functions (e.g., getdirentries64()), concealing 6 targeted files and spoofing directory contents returned to userspace tools for stealth persistence.	

Skills

Languages: JavaScript, TypeScript, HTML, CSS, SQL, C/C++, Python, Java, Rust, Shell, Bash.
Technologies: ReactJS, NextJS, VanillaJS, Node.js, Express, PostgreSQL, REST APIs, CI/CD, AWS (EC2, S3), POSIX, pthreads, mmap, perf, Valgrind, GDB, eBPF, Kernel Programming, TCP/IP, DNS, Sockets, Linux/Unix